Mark Tinega

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PROFESSIONAL SUMMARY

Software and AI/ML Engineer with 5+ years building production systems across the stack: ML pipelines, LLM-powered applications, backend APIs, and the infrastructure that keeps them running reliably at scale. Comfortable moving between model work, backend engineering, and infrastructure depending on what the problem needs.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C#/.NET, Bash

ML & AI: LangGraph, LiteLLM, Anthropic Claude API, OpenAI API, Ollama, vLLM, Hugging Face, PyTorch, scikit-learn, XGBoost, LightGBM, IsolationForest, RAG pipelines, pgvector, prompt engineering, model-as-a-judge evaluation, MCP tool integrations

MLOps & Platforms: MLRun, Iguazio, Kubeflow Pipelines (KFP), MLflow, Kedro, model versioning, feature stores, batch and real-time inference, A/B experiment tracking

Data Engineering: PySpark, Apache Kafka, ETL pipelines, PostgreSQL, pgvector, Redis, MongoDB, OpenSearch, feature engineering, embedding pipelines

Backend: FastAPI, Django/DRF, Node.js/NestJS, REST API design, async programming, OAuth2/JWT, Alembic migrations, Microsoft SQL Server

Infrastructure: Kubernetes (EKS), Docker, AWS (EC2, S3, Lambda, SageMaker, Bedrock, API Gateway, ECR, RDS), Helm, Terraform, Nuclio, Jenkins, GitHub Actions

Observability: Prometheus, Grafana, structured logging, alerting pipelines, token consumption tracking, latency distributions

Frontend: React, TypeScript, HTML/CSS, REST integrations

WORK EXPERIENCE

Machine Learning Engineer | Safaricom PLC, Nairobi, Kenya *Mar 2025 – Present*

Kenya's largest telco, 50M+ customers. Owns AI platform operations, production ML pipelines, LLM systems, and ML serving infrastructure on Iguazio/MLRun/EKS.

- Own the AI platform (Iguazio/MLRun) running all production ML workloads: model versioning, scheduled pipeline execution via Kubeflow Pipelines, and EKS cluster operations for inference in AWS eu-west-1; inference infrastructure serves 5,000 TPS across model families.
- Built and operate a multi-provider LLM gateway (LiteLLM, AWS API Gateway) as the single control plane for all LLM access across the organisation: unified routing across Anthropic, OpenAI, and AWS Bedrock, per-team spend tracking, rate limiting, and a single InfoSec audit point.
- Built and maintain production ML classification and risk-rating pipelines (Kedro, IsolationForest, MLRun) with automated scheduling, monitoring metrics jobs, and score exports to PostgreSQL synced to AWS RDS.
- Built a multi-agent LLM system (LangGraph supervisor-worker pattern, Anthropic Claude API) deployed via Nuclio on Kubernetes that assists 1,000+ call centre agents in real time; led model evaluation using a model-as-a-judge framework and shipped a 22% quality improvement over two quarters.
- Deploy and serve open-weight models (Qwen, vLLM) on internal GPU infrastructure for data-sovereign workloads; owns workload classification, capacity planning, and inference performance tuning.
- Built and maintains a production RAG pipeline (pgvector, PostgreSQL): embedding ingestion, chunking, retrieval re-ranking, and a feedback loop that feeds flagged responses back into prompt tuning cycles.
- Build full-stack Python applications (FastAPI, Django) that expose ML models and AI systems to end users: REST APIs, async inference endpoints, React/TypeScript frontends, and real-time interfaces for model-powered features.
- Instrumented ML and LLM systems with Prometheus and Grafana covering token consumption, latency distributions, error rates, and anomaly alerts; resolved a Kafka consumer pool exhaustion incident via load profiling and backpressure tuning.

Software Engineer | Huawei Technologies, Nairobi, Kenya *Dec 2023 – Dec 2024*

Global technology company. Backend services and solution architecture for fintech products used by millions of Kenyans.

- Designed and delivered Python/FastAPI backend services for cloud-native fintech integrations: OAuth2 authentication flows, async data pipelines, and REST API contracts across multiple payment and lending products.

- Optimised a Microsoft SQL Server reconciliation query from 45 seconds to under 2 seconds via non-clustered index design and T-SQL restructuring, unblocking a daily finance reporting workflow.
- Delivered solution architecture for enterprise clients, translating technical trade-offs into investment recommendations for executive stakeholders; contributed to Kubernetes CI/CD automation and containerised deployments.
- Provisioned and operated VM and Huawei Cloud infrastructure for fintech workloads as Software Service Engineer (Jul 2023 - Dec 2023), covering infrastructure-as-code and service reliability for payment backends.

Senior ML & Backend Engineer | Fiverr Pro (Freelance), Remote *Nov 2019 – Present*

Independent consulting practice. Clients across the US, UK, and East Africa.

- Built a production RAG pipeline for a US client: ingested, chunked, and embedded large document corpora into a vector store, then exposed a retrieval-augmented Q&A interface via FastAPI with async inference and structured output parsing.
- Built a lead scoring and routing pipeline combining PostgreSQL, Python ETL, and a classification model, reducing manual triage time by 60% for a US marketing client.
- Built a real-time sports data ingestion, prediction, and automation platform: end-to-end ML pipeline from data ingestion through model inference to execution layer.
- Delivered ML model integrations (OpenAI, Hugging Face) with async inference, retry logic with exponential backoff, and structured output parsing for client automation workflows.
- Built a document parsing and NLP classification pipeline to extract structured data from legal communication records and deliver a clean reporting interface.

Backend Engineer | Zuyu, London, UK (Remote) *2022 – 2023*

Early-stage startup building payment and logistics products for East Africa.

- Built Django/DRF REST APIs and PostgreSQL schemas from scratch for a multi-tenant payment platform; owned backend releases end-to-end including schema migrations, deployment, and post-deploy monitoring.
- Implemented OAuth2 and JWT authentication infrastructure using NestJS: token rotation, refresh flows, rate limiting, and audit logging for a multi-tenant platform.
- Designed core payment and logistics API contracts in collaboration with the founding team, translating product requirements into backend architecture decisions.
- Set up automated testing and deployment pipelines, reducing release overhead and enabling faster iteration with fewer regressions.

Data Analyst Intern | SHOFCO, Nairobi, Kenya *2021 – 2022*

Social enterprise serving urban communities across Kenya.

- Built Python data pipelines and ETL workflows consolidating programme monitoring data from 10 community centres into a central reporting system; reduced the reporting cycle from two weeks to two days.
- Designed reporting dashboards giving programme managers real-time visibility into beneficiary outcomes across health, education, and economic programmes.
- Automated manual data collection processes previously handled in spreadsheets, improving data quality and reducing staff reporting time by an estimated 70%.
- Worked directly with non-technical programme staff to translate operational data needs into engineering requirements.

EDUCATION

BSc Mathematics and Computer Science | Kenyatta University, Nairobi, Kenya *2018 – 2023*

Relevant coursework: Linear Algebra, Calculus and Real Analysis, Probability and Statistics, Discrete Mathematics, Data Structures and Algorithms, Database Systems, Operating Systems, Computer Networks, Numerical Methods, Machine Learning, Artificial Intelligence, Software Engineering.

CERTIFICATIONS

- AWS Certified AI Practitioner (2026)
- Advanced Learning Algorithms, Stanford University / Coursera (Nov 2025)
- Supervised Machine Learning: Regression and Classification, Stanford University / Coursera (Sep 2025)
- IBM AI Engineering, Coursera
- PCAP, Python Institute (2022)
- Cisco Certified Network Associate (CCNA)